

CS-306-MJ-P: Core Java and Web Technology-I

Slip-wise Solutions (T.Y.B.Sc. CS, SPPU, Sem V)

One simplest question is solved from each of the 30 slips. Each question carries 15 marks, so a fully correct, working answer clears the 14-mark passing target on its own. Tip: whichever question you attempt, make sure it runs and shows output in front of the examiner, and keep the file names as given.

SLIP 1 — Solve Q1 (15 marks)

Question: Write a Java program to print the sum of elements of the array. Also display array elements in ascending order.

Language: Java **File:** ArraySum.java

Run: javac ArraySum.java | java ArraySum

```
import java.util.*;

public class ArraySum {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter size of array: ");
        int n = sc.nextInt();
        int[] a = new int[n];
        int sum = 0;
        System.out.println("Enter " + n + " elements:");
        for (int i = 0; i < n; i++) {
            a[i] = sc.nextInt();
            sum += a[i];
        }
        System.out.println("Sum of elements = " + sum);
        Arrays.sort(a);
        System.out.println("Array in ascending order:");
        for (int x : a)
            System.out.print(x + " ");
        System.out.println();
    }
}
```

SLIP 2 — Solve Q1 (15 marks)

Question: Write a Java Program to Display Armstrong Numbers Between range. Accept range from user.

Language: Java **File:** Armstrong.java

Run: javac Armstrong.java | java Armstrong

```
import java.util.*;

public class Armstrong {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter start of range: ");
        int start = sc.nextInt();
        System.out.print("Enter end of range: ");
        int end = sc.nextInt();
        System.out.println("Armstrong numbers between " + start + " and " + end + ":");
        for (int i = start; i <= end; i++) {
            int temp = i, sum = 0;
            int digits = String.valueOf(i).length();
            while (temp > 0) {
                int d = temp % 10;
                sum += (int) Math.pow(d, digits);
                temp /= 10;
            }
            if (sum == i)
                System.out.println(i);
        }
    }
}
```

SLIP 3 — Solve Q2 (15 marks)

Question: Create a responsive webpage using CSS Flexbox for a college homepage containing header, navigation, content and footer. Use appropriate Flexbox alignment and distribution properties.

Language: HTML + CSS **File:** college.html

Run: Open college.html in a browser (resize the window to see it respond)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>College Homepage</title>
<style>
* { margin: 0; padding: 0; box-sizing: border-box; }
body { font-family: Arial, sans-serif; min-height: 100vh;
      display: flex; flex-direction: column; }
header { background: #1a3c6e; color: #fff; padding: 20px; text-align: center; }
nav { background: #2c5aa0; display: flex; justify-content: space-around;
      flex-wrap: wrap; }
nav a { color: #fff; text-decoration: none; padding: 14px 20px; }
nav a:hover { background: #1a3c6e; }
main { flex: 1; display: flex; flex-wrap: wrap; justify-content: center;
      align-items: flex-start; gap: 20px; padding: 20px; }
.card { flex: 1 1 250px; background: #eef3fa; padding: 20px; border-radius: 8px; }
footer { background: #222; color: #fff; text-align: center; padding: 15px; }
@media (max-width: 600px) { nav { flex-direction: column; align-items: center; } }
</style>
</head>
<body>
<header>
<h1>ABC College of Science</h1>
<p>Excellence in Education</p>
</header>
<nav>
<a href="#">Home</a>
<a href="#">About</a>
<a href="#">Courses</a>
<a href="#">Admission</a>
<a href="#">Contact</a>
</nav>
<main>
<div class="card"><h2>About Us</h2>
  <p>We offer quality education in science and computer studies.</p></div>
<div class="card"><h2>Courses</h2>
  <p>B.Sc. Computer Science, B.Sc. Electronics, BBA (CA).</p></div>
<div class="card"><h2>Notices</h2>
  <p>Admissions are open for the new academic year.</p></div>
</main>
<footer>&copy; 2026 ABC College. All rights reserved.</footer>
</body>
</html>
```

SLIP 4 — Solve Q2 (15 marks)

Question: Create a webpage containing a heading and a button. When the button is clicked, change the heading text from "Hello! Welcome" to "Text Changed Successfully!".

Language: HTML + JavaScript **File:** change_text.html

Run: Open change_text.html in a browser and click the button

```
<!DOCTYPE html>
<html>
<head>
<title>Change Text</title>
</head>
<body>
<h1 id="heading">Hello! Welcome</h1>
<button onclick="changeText()">Change Text</button>

<script>
function changeText() {
  document.getElementById("heading").innerText = "Text Changed Successfully!";
}
</script>
</body>
```

</html>

SLIP 5 — Solve Q2 (15 marks)

Question: Create an image gallery using HTML5 and CSS3 with hover effects, responsive images and Flexbox/Grid layout.

Language: HTML + CSS **File:** gallery.html

Run: Put 6 images named img1.jpg ... img6.jpg in the same folder as gallery.html, then open it

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<title>Image Gallery</title>
<style>
  * { box-sizing: border-box; margin: 0; padding: 0; }
  body { font-family: Arial, sans-serif; background: #f4f4f4; padding: 20px; }
  h1 { text-align: center; margin-bottom: 20px; }
  .gallery { display: flex; flex-wrap: wrap; gap: 15px; justify-content: center; }
  .item { flex: 1 1 250px; max-width: 300px; overflow: hidden;
    border-radius: 8px; box-shadow: 0 2px 6px rgba(0,0,0,0.3); }
  .item img { width: 100%; height: 200px; object-fit: cover; display: block;
    transition: transform 0.4s, filter 0.4s; }
  .item:hover img { transform: scale(1.15); filter: brightness(1.15); }
</style>
</head>
<body>
  <h1>My Image Gallery</h1>
  <div class="gallery">
    <div class="item"></div>
    <div class="item"></div>
    <div class="item"></div>
    <div class="item"></div>
    <div class="item"></div>
    <div class="item"></div>
  </div>
</body>
</html>
```

SLIP 6 — Solve Q2 (15 marks)

Question: Create a webpage containing a button and link. Change their appearance when the mouse pointer moves over them. (Use :hover Pseudo-Class)

Language: HTML + CSS **File:** hover.html

Run: Open hover.html in a browser and move the mouse over the button and link

```
<!DOCTYPE html>
<html>
<head>
<title>Hover Effect</title>
<style>
  body { font-family: Arial, sans-serif; margin: 40px; }
  button { padding: 10px 25px; background: #007bff; color: #fff; border: none;
    font-size: 16px; cursor: pointer; transition: 0.3s; }
  button:hover { background: #ff5722; transform: scale(1.1); }
  a { font-size: 18px; color: blue; text-decoration: none; margin-left: 20px; }
  a:hover { color: #fff; background: green; text-decoration: underline; padding: 5px; }
</style>
</head>
<body>
  <button>Click Me</button>
  <a href="https://www.google.com">Visit Google</a>
</body>
</html>
```

SLIP 7 — Solve Q2 (15 marks)

Question: Create an unordered list of Programming Language names and apply different styles to the first and last list items. (Use `:first-child` and `:last-child`)

Language: HTML + CSS **File:** list.html

Run: Open list.html in a browser

```
<!DOCTYPE html>
<html>
<head>
<title>Programming Languages</title>
<style>
  ul { font-size: 20px; }
  li:first-child { color: white; background: green; font-weight: bold; }
  li:last-child { color: red; font-style: italic; font-size: 24px; }
</style>
</head>
<body>
  <h2>Programming Languages</h2>
  <ul>
    <li>C</li>
    <li>C++</li>
    <li>Java</li>
    <li>Python</li>
    <li>JavaScript</li>
  </ul>
</body>
</html>
```

SLIP 8 — Solve Q2 (15 marks)

Question: Create a webpage using JavaScript DOM manipulation and event handling with an image and a button. When the button is clicked, replace the displayed image with another image.

Language: HTML + JavaScript **File:** image_change.html

Run: Keep img1.jpg and img2.jpg in the same folder as image_change.html, then open it

```
<!DOCTYPE html>
<html>
<head>
  <title>Change Image</title>
</head>
<body>
  
  <br><br>
  <button id="btn">Change Image</button>

  <script>
    document.getElementById("btn").addEventListener("click", function () {
      var img = document.getElementById("pic");
      if (img.src.includes("img1.jpg"))
        img.src = "img2.jpg";
      else
        img.src = "img1.jpg";
    });
  </script>
</body>
</html>
```

SLIP 9 — Solve Q2 (15 marks)

Question: Write a JavaScript program to demonstrate Arrow Functions for addition, subtraction, multiplication and division.

Language: JavaScript **File:** arrow.js

Run: node arrow.js

```
const add = (a, b) => a + b;
const subtract = (a, b) => a - b;
const multiply = (a, b) => a * b;
const divide = (a, b) => (b === 0 ? "Cannot divide by zero" : a / b);

let x = 20, y = 5;
console.log(`Addition      : ${x} + ${y} = ${add(x, y)}`);
console.log(`Subtraction   : ${x} - ${y} = ${subtract(x, y)}`);
console.log(`Multiplication : ${x} * ${y} = ${multiply(x, y)}`);
console.log(`Division      : ${x} / ${y} = ${divide(x, y)}`);
```

SLIP 10 — Solve Q2 (15 marks)

Question: Write a JavaScript program to store student information and marks in variables and generate a formatted student report using template literals.

Language: JavaScript **File:** report.js

Run: node report.js

```
const name = "Rahul Patil";
const rollNo = 34;
const className = "T.Y.B.Sc. Computer Science";
const m1 = 78, m2 = 85, m3 = 90;

const total = m1 + m2 + m3;
const percentage = (total / 300 * 100).toFixed(2);

const report = `
===== STUDENT REPORT =====
Name      : ${name}
Roll No   : ${rollNo}
Class     : ${className}
-----
Subject 1 : ${m1}
Subject 2 : ${m2}
Subject 3 : ${m3}
-----
Total      : ${total} / 300
Percentage : ${percentage}%
Result     : ${percentage >= 40 ? "PASS" : "FAIL"}
=====`;

console.log(report);
```

SLIP 11 — Solve Q2 (15 marks)

Question: Write a JavaScript program using an arrow function to calculate the total and average marks of three subjects. Display the result using template literals.

Language: JavaScript **File:** marks.js

Run: node marks.js

```
const calculate = (m1, m2, m3) => {
  const total = m1 + m2 + m3;
  const average = total / 3;
  return { total, average };
};

const m1 = 70, m2 = 80, m3 = 90;
const result = calculate(m1, m2, m3);

console.log(`Marks      : ${m1}, ${m2}, ${m3}`);
console.log(`Total      : ${result.total}`);
console.log(`Average   : ${result.average.toFixed(2)}`);
```

SLIP 12 — Solve Q2 (15 marks)

Question: Write a JavaScript program using an arrow function to calculate the total price of a product. Display the bill using template literals and string interpolation.

Language: JavaScript **File:** bill.js

Run: node bill.js

```
const totalPrice = (price, qty, gstPercent) => {
  const amount = price * qty;
  const gst = amount * gstPercent / 100;
  return { amount, gst, total: amount + gst };
};

const product = "Wireless Mouse";
const price = 500;
const qty = 3;
const gstPercent = 18;

const bill = totalPrice(price, qty, gstPercent);

console.log(`
***** BILL *****
Product   : ${product}
Price     : Rs. ${price}
Quantity  : ${qty}
Amount    : Rs. ${bill.amount}
GST (${gstPercent}%): Rs. ${bill.gst}
TOTAL     : Rs. ${bill.total}
*****`);
```

SLIP 13 — Solve Q2 (15 marks)

Question: Write a JavaScript program to swap two numbers without using a third variable using array destructuring.

Language: JavaScript **File:** swap.js

Run: node swap.js

```
let a = 10;
let b = 20;

console.log(`Before swapping: a = ${a}, b = ${b}`);

[a, b] = [b, a];

console.log(`After swapping : a = ${a}, b = ${b}`);
```

SLIP 14 — Solve Q2 (15 marks)

Question: Create an object containing employee details such as name, department, and salary. Use object destructuring to extract and display the values.

Language: JavaScript **File:** employee.js

Run: node employee.js

```
const employee = {
  name: "Amit Sharma",
  department: "IT",
  salary: 45000
};

const { name, department, salary } = employee;

console.log("Employee Details");
console.log("Name      : " + name);
console.log("Department : " + department);
console.log("Salary     : " + salary);
```

SLIP 15 — Solve Q2 (15 marks)

Question: Write a JavaScript program to create a copy of an existing array using the spread operator and add a new element to the copied array.

Language: JavaScript **File:** spread.js

Run: node spread.js

```
const original = [10, 20, 30, 40];
const copy = [...original];
copy.push(50);

console.log("Original array :", original);
console.log("Copied array  :", copy);
```

SLIP 16 — Solve Q2 (15 marks)

Question: Write a JavaScript program using a rest parameter to accept any number of values and calculate their sum.

Language: JavaScript **File:** rest.js

Run: node rest.js

```
function sum(...numbers) {
  let total = 0;
  for (const n of numbers) {
    total += n;
  }
  return total;
}

console.log("Sum of 1, 2, 3          = " + sum(1, 2, 3));
console.log("Sum of 10, 20, 30, 40  = " + sum(10, 20, 30, 40));
console.log("Sum of 5                = " + sum(5));
```

SLIP 17 — Solve Q2 (15 marks)

Question: Create a JavaScript module containing arithmetic operation functions. Export the functions and import them into another JavaScript file to perform all operations.

Language: JavaScript (ES Module) **File:** mathops.mjs + main.mjs

Run: node main.mjs (both files in the same folder; .mjs extension enables import/export)

```
// ----- File 1: mathops.mjs -----
export const add = (a, b) => a + b;
export const subtract = (a, b) => a - b;
export const multiply = (a, b) => a * b;
export const divide = (a, b) => (b === 0 ? "Cannot divide by zero" : a / b);

// ----- File 2: main.mjs -----
import { add, subtract, multiply, divide } from "./mathops.mjs";

const x = 20, y = 4;
console.log("Addition      :", add(x, y));
console.log("Subtraction   :", subtract(x, y));
console.log("Multiplication :", multiply(x, y));
console.log("Division       :", divide(x, y));
```

SLIP 18 — Solve Q2 (15 marks)

Question: Create a Node.js HTTP server that sends an appropriate response to a client request.

Language: Node.js **File:** server.js

Run: node server.js then open http://localhost:3000 in a browser

```
const http = require("http");

const server = http.createServer((req, res) => {
  if (req.url === "/" ) {
    res.writeHead(200, { "Content-Type": "text/html" });
    res.end("<h1>Hello from Node.js Server!</h1>");
  } else if (req.url === "/about") {
    res.writeHead(200, { "Content-Type": "text/plain" });
    res.end("This is the About page.");
  } else {
    res.writeHead(404, { "Content-Type": "text/plain" });
    res.end("404 - Page Not Found");
  }
});

server.listen(3000, () => {
  console.log("Server running at http://localhost:3000");
});
```

SLIP 19 — Solve Q2 (15 marks)

Question: Write a JavaScript program using async/await to simulate user login. If the username and password are correct, display a success message; otherwise, handle the error using try-catch.

Language: JavaScript **File:** login.js

Run: node login.js

```
function login(username, password) {
  return new Promise((resolve, reject) => {
    setTimeout(() => {
      if (username === "admin" && password === "1234")
        resolve("Login successful. Welcome, " + username + "!");
      else
        reject(new Error("Invalid username or password"));
    }, 1000);
  });
}

async function main() {
  try {
    const msg = await login("admin", "1234");
    console.log(msg);
  } catch (err) {
    console.log("Error: " + err.message);
  }

  try {
    const msg = await login("admin", "wrong");
    console.log(msg);
  } catch (err) {
    console.log("Error: " + err.message);
  }
}

main();
```

SLIP 20 — Solve Q2 (15 marks)

Question: Write a JavaScript program using a Promise to divide two numbers. If the denominator is zero, reject the Promise and handle the error.

Language: JavaScript **File:** divide.js

Run: node divide.js

```
function divide(a, b) {
  return new Promise((resolve, reject) => {
    if (b === 0)
      reject("Error: Denominator cannot be zero");
    else
      resolve(a / b);
  });
}

divide(20, 4)
  .then((result) => console.log("20 / 4 = " + result))
  .catch((err) => console.log(err));

divide(10, 0)
  .then((result) => console.log("10 / 0 = " + result))
  .catch((err) => console.log(err));
```

SLIP 21 — Solve Q2 (15 marks)

Question: Write a Node.js program using the path module to perform operations such as joining, resolving and extracting file/directory information.

Language: Node.js **File:** pathdemo.js

Run: node pathdemo.js

```
const path = require("path");

const filePath = "/home/student/docs/report.txt";

console.log("Join      :", path.join("/home", "student", "docs", "..", "notes.txt"));
console.log("Resolve   :", path.resolve("docs", "file.txt"));
console.log("Normalize  :", path.normalize("/home//student/./docs/./a.txt"));
console.log("Dirname    :", path.dirname(filePath));
console.log("Basename   :", path.basename(filePath));
console.log("Base (no ext):", path.basename(filePath, ".txt"));
console.log("Extension  :", path.extname(filePath));
console.log("Absolute?   :", path.isAbsolute(filePath));
console.log("Parse      :", path.parse(filePath));
```

SLIP 22 — Solve Q2 (15 marks)

Question: Write a Node.js program using the fs module to perform file operations asynchronously.

Language: Node.js **File:** asyncfs.js

Run: node asyncfs.js

```
const fs = require("fs");

// 1. Create and write
fs.writeFile("demo.txt", "Hello, this is async file handling.\n", (err) => {
  if (err) return console.log("Write error:", err.message);
  console.log("File created and written.");
});

// 2. Append
fs.appendFile("demo.txt", "This line is appended.\n", (err) => {
  if (err) return console.log("Append error:", err.message);
  console.log("Data appended.");
});

// 3. Read
fs.readFile("demo.txt", "utf8", (err, data) => {
  if (err) return console.log("Read error:", err.message);
  console.log("File contents:\n" + data);
});

// 4. Delete
fs.unlink("demo.txt", (err) => {
  if (err) return console.log("Delete error:", err.message);
  console.log("File deleted.");
});
});
```

```
});  
});
```

SLIP 23 — Solve Q2 (15 marks)

Question: Write a Node.js program to create, write, read and append data to a file using the fs module.

Language: Node.js **File:** fileops.js

Run: node fileops.js

```
const fs = require("fs");  
const file = "student.txt";  
  
// Create and write  
fs.writeFileSync(file, "Name: Rahul\nClass: TYBSc CS\n");  
console.log("File created and data written.");  
  
// Read  
console.log("\nContents after write:");  
console.log(fs.readFileSync(file, "utf8"));  
  
// Append  
fs.appendFileSync(file, "City: Nashik\n");  
console.log("Data appended.");  
  
// Read again  
console.log("\nContents after append:");  
console.log(fs.readFileSync(file, "utf8"));
```

SLIP 24 — Solve Q1 (15 marks)

Question: Create an abstract class Shape with methods area & volume. Derive a class Cylinder (radius, height). Calculate area and volume.

Language: Java **File:** CylinderDemo.java

Run: javac CylinderDemo.java | java CylinderDemo

```
import java.util.*;  
  
abstract class Shape {  
    abstract double area();  
    abstract double volume();  
}  
  
class Cylinder extends Shape {  
    double radius, height;  
  
    Cylinder(double radius, double height) {  
        this.radius = radius;  
        this.height = height;  
    }  
  
    double area() { // total surface area  
        return 2 * Math.PI * radius * (radius + height);  
    }  
  
    double volume() {  
        return Math.PI * radius * radius * height;  
    }  
}  
  
public class CylinderDemo {  
    public static void main(String[] args) {  
        Scanner sc = new Scanner(System.in);  
        System.out.print("Enter radius: ");  
        double r = sc.nextDouble();  
        System.out.print("Enter height: ");  
        double h = sc.nextDouble();  
  
        Cylinder c = new Cylinder(r, h);  
        System.out.printf("Surface Area = %.2f\n", c.area());  
        System.out.printf("Volume = %.2f\n", c.volume());  
    }  
}
```

SLIP 25 — Solve Q1 (15 marks)

Question: Write a Java program that defines an interface Calculator containing methods add() and subtract() and implements it in class SimpleCalc.

Language: Java **File:** CalcDemo.java

Run: javac CalcDemo.java | java CalcDemo

```
import java.util.*;

interface Calculator {
    int add(int a, int b);
    int subtract(int a, int b);
}

class SimpleCalc implements Calculator {
    public int add(int a, int b) {
        return a + b;
    }

    public int subtract(int a, int b) {
        return a - b;
    }
}

public class CalcDemo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter first number: ");
        int x = sc.nextInt();
        System.out.print("Enter second number: ");
        int y = sc.nextInt();

        Calculator c = new SimpleCalc();
        System.out.println("Addition    = " + c.add(x, y));
        System.out.println("Subtraction = " + c.subtract(x, y));
    }
}
```

SLIP 26 — Solve Q1 (15 marks)

Question: Define a class Student with data members student_name, student_rollno, total_lectures, attended_lectures. Create an object of Student. If attendance percentage is less than 75%, throw a user-defined exception "Student is Not Eligible for Exam", otherwise display student details.

Language: Java **File:** StudentDemo.java

Run: javac StudentDemo.java | java StudentDemo

```
import java.util.*;

class NotEligibleException extends Exception {
    NotEligibleException(String msg) {
        super(msg);
    }
}

class Student {
    String student_name;
    int student_rollno, total_lectures, attended_lectures;

    Student(String name, int roll, int total, int attended) {
        student_name = name;
        student_rollno = roll;
        total_lectures = total;
        attended_lectures = attended;
    }

    void checkEligibility() throws NotEligibleException {
        double percent = (attended_lectures * 100.0) / total_lectures;
        if (percent < 75)
            throw new NotEligibleException("Student is Not Eligible for Exam");
    }

    void display() {
        double percent = (attended_lectures * 100.0) / total_lectures;
        System.out.println("Name          : " + student_name);
    }
}
```

```

        System.out.println("Roll No      : " + student_rollno);
        System.out.println("Attendance : " + percent + "%");
    }
}

public class StudentDemo {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter name: ");
        String name = sc.nextLine();
        System.out.print("Enter roll no: ");
        int roll = sc.nextInt();
        System.out.print("Enter total lectures: ");
        int total = sc.nextInt();
        System.out.print("Enter attended lectures: ");
        int attended = sc.nextInt();

        Student s = new Student(name, roll, total, attended);
        try {
            s.checkEligibility();
            s.display();
        } catch (NotEligibleException e) {
            System.out.println(e.getMessage());
        }
    }
}

```

SLIP 27 — Solve Q2 (15 marks)

Question: Write a Node.js program to read a JSON file, modify selected properties of the JSON object, and write the updated JSON data back to the file using the fs module.

Language: Node.js **File:** updatejson.js

Run: node updatejson.js

```

const fs = require("fs");
const file = "data.json";

// Create a sample JSON file if it does not exist
if (!fs.existsSync(file)) {
    const sample = { id: 1, name: "Amit", city: "Pune", marks: 70 };
    fs.writeFileSync(file, JSON.stringify(sample, null, 2));
}

fs.readFile(file, "utf8", (err, data) => {
    if (err) return console.log("Read error:", err.message);

    const obj = JSON.parse(data);
    console.log("Original:", obj);

    // Modify selected properties
    obj.name = "Rahul";
    obj.city = "Nashik";
    obj.marks = 85;

    fs.writeFile(file, JSON.stringify(obj, null, 2), (err) => {
        if (err) return console.log("Write error:", err.message);
        console.log("Updated JSON written to " + file);
        console.log("Updated:", obj);
    });
});

```

SLIP 28 — Solve Q2 (15 marks)

Question: Create a Node.js program to list files in a specified directory, filter files based on their extensions (such as .txt, .json and .js), and display the matching file names.

Language: Node.js **File:** filter.js

Run: node filter.js [folder-path] (uses current folder if no path is given)

```
const fs = require("fs");
const path = require("path");

const dir = process.argv[2] || ".";
const extensions = [".txt", ".json", ".js"];

fs.readdir(dir, (err, files) => {
  if (err) return console.log("Error:", err.message);

  const matched = files.filter((f) =>
    extensions.includes(path.extname(f).toLowerCase())
  );

  if (matched.length === 0) {
    console.log("No matching files found.");
  } else {
    console.log("Matching files in " + dir + ":");
    matched.forEach((f) => console.log(f));
  }
});
```

SLIP 29 — Solve Q2 (15 marks)

Question: Write a Node.js program to demonstrate file statistics using fs.stat(), including file size, file type, creation/access/modification information, and whether the path represents a file or directory.

Language: Node.js **File:** stats.js

Run: node stats.js [path] (checks the script itself if no path is given)

```
const fs = require("fs");

const target = process.argv[2] || __filename;

fs.stat(target, (err, stats) => {
  if (err) return console.log("Error:", err.message);

  console.log("Path          :", target);
  console.log("Size (bytes)   :", stats.size);
  console.log("Is File         :", stats.isFile());
  console.log("Is Directory    :", stats.isDirectory());
  console.log("Created         :", stats.birthtime);
  console.log("Last Accessed  :", stats.atime);
  console.log("Last Modified  :", stats.mtime);
  console.log("Type           :",
    stats.isFile() ? "File" : stats.isDirectory() ? "Directory" : "Other");
});
```

SLIP 30 — Solve Q1 (15 marks)

Question: Define a class MyNumber having one private integer data member. Write a default constructor to initialize it to 0 and another constructor to initialize it to a value. Write methods isNegative, isPositive, isOdd, isEven. Use command line argument to pass a value to the object and perform the above operations.

Language: Java **File:** MyNumberDemo.java

Run: javac MyNumberDemo.java | java MyNumberDemo 25

```
class MyNumber {
  private int n;

  MyNumber() {
    n = 0;
  }

  MyNumber(int n) {
    this.n = n;
  }

  boolean isNegative() { return n < 0; }
```

```
    boolean isPositive() { return n > 0; }
    boolean isEven()     { return n % 2 == 0; }
    boolean isOdd()      { return n % 2 != 0; }
}

public class MyNumberDemo {
    public static void main(String[] args) {
        if (args.length == 0) {
            System.out.println("Usage: java MyNumberDemo <number>");
            return;
        }
        int value = Integer.parseInt(args[0]);
        MyNumber m = new MyNumber(value);

        System.out.println("Number      : " + value);
        System.out.println("Negative    : " + m.isNegative());
        System.out.println("Positive   : " + m.isPositive());
        System.out.println("Even       : " + m.isEven());
        System.out.println("Odd        : " + m.isOdd());
    }
}
```
